CS201 Lab 7

1. The equation of a forced-damped oscillator is given by,

$$\frac{d^2x}{dt^2} + 2\beta\frac{dx}{dt} + \omega_0^2 x = F_0\cos(\omega t)$$

   (a) Choose numerical values $F_0 = 1, \omega_0 = 2.5$ and $\beta = 0.5$ units. Put $\omega = 0.3$ and graph the transient and the oscillatory behaviour by numerically solving the ODE up to $t = 100$ seconds. You may choose initial conditions, $x(0) = 0, \dot{x}(0) = 1$.

   (b) By adjusting ω, try to observe the phenomenon of resonance (maximization of amplitude). Plot the results of t vs $x(t)$ for three values of $\omega = \omega_0$, less than ω_0 and greater than ω_0.

   (c) Try to modify your code to plot maximum amplitude vs driving frequency. Does this graph show a maximum? If so, at what value of the driving frequency?

2. Consider the oscillator,

$$\ddot{x} + x = \cos(\omega t), x(0) = 0, x'(0) = 0$$

   In the above equation, plot t vs $x(t)$ up to $t = 20\pi$ and explore how the graph changes with increasing ω, from small to large. Present results for three values of ω. Try plotting for $\omega \approx \omega_0$, where ω_0 is the natural frequency. Can you explain the results theoretically (you may need the analytical solution for this).